THE EARTH'S CRUST, ROCKS / INDIA GEOLOGICAL STRUCTURE

CRUST, LITHOSPHERE AND THE → IGNEOUS, SEDIMENTARY AND METAMORPHIC → FROM CONTINENTAL DRIFT TO → INDIAN PLATE JOURNEY AND → INDIAN ROCK SYSTEMS, BASINS → HIMALAYA, FORELAND, COASTS, ISLANDS AND STRUCTURAL

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW ● SUBORDINATE EXTRA ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • source generation g14 and all prior artifacts unchanged

00

STAGE 00 CRUST, LITHOSPHERE AND THE ROCK-CYCLE NETWORK

CRUST, LITHOSPHERE AND THE ROCK-CYCLE NETWORK · EARTH MATERIAL SYSTEM MINERAL → ROCK → LITHOLOGY → PETROLOGY → GEOLOGY · COMPOSITIONAL SHELL · CRUST + RIGID UPPERMOST MANTLE · WEAKER MECHANICAL LAYER BELOW · ROCK-CYCLE NETWORK MAGMA →CRYSTALLISATION→ IGNEOUS WEATHERING + TRANSPORT <·····> MELTING → METAMORPHIC <·· SEDIMENTARY HEAT + PRESSSURE · THE CYCLE IS A NETWORK

EARTH MATERIAL SYSTEM mineral → rock → lithology → petrology → geology → Crust = compositional shell → Lithosphere = crust + rigid uppermost mantle → Asthenosphere = weaker mechanical layer below → ROCK-CYCLE NETWORK magma →crystallisation→ IGNEOUS weathering + transport

CORE CLAIM
- EARTH MATERIAL SYSTEM mineral → rock → lithology → petrology → geology
- Crust = compositional shell
- Lithosphere = crust + rigid uppermost mantle

MECHANISM
- Asthenosphere = weaker mechanical layer below
- ROCK-CYCLE NETWORK: magma →crystallisation→ IGNEOUS weathering + transport
- <·····> melting ·· METAMORPHIC <·· SEDIMENTARY heat + pressure compaction + cementation uplift and exposure reconnect all paths

CONSEQUENCE / LIMIT
- The cycle is a network: no rock family has one compulsory next stage.
- MUST REMEMBER: Crustal analysis connects seismic evidence, compositional and rheological layers, rock cycle, plate boundaries and
- Wilson-cycle logic to India's cratons, mobile belts, Gondwana basins, Deccan volcanics, Himalaya and Indo-Gangetic foredeep.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: The crust is Earth's outer compositional shell, whereas the lithosphere is the stronger mechanical layer comprising the crust plus the rigid uppermost mantle.

01

STAGE 01 IGNEOUS, SEDIMENTARY AND METAMORPHIC DIAGNOSTIC MATRIX

IGNEOUS, SEDIMENTARY AND METAMORPHIC DIAGNOSTIC MATRIX · FIELD CLUES · MAGMA/LAVA COOLING · CRYSTALS, GLASS, VESICLES INTRUSIVE · SLOW COOLING · COARSE TEXTURE EXTRUSIVE · FAST COOLING · FINE OR GLASSY TEXTURE

FAMILY	PROCESS	FIELD CLUES
Sedimentary	deposit + lithify	beds, fossils, ripples
granite-rhyolite + felsic	diorite-andesite + intermediate gabbro-basalt + mafic	
FIREWALLS bedding is not foliation	loess and moraine are deposits until lithified porosity is void space	permeability is connected flow capacity

CORE CLAIM
- FIELD CLUES
- magma/lava cooling
- crystals, glass, vesicles intrusive
- slow cooling
- coarse texture extrusive

MECHANISM
- deposit + lithify
- beds, fossils, ripples
- solid-state change
- grade, new minerals foliated
- differential stress

CONSEQUENCE / LIMIT
- granite-rhyolite + felsic
- diorite-andesite + intermediate gabbro-basalt + mafic
- FIREWALLS bedding is not foliation
- loess and moraine are deposits until lithified porosity is void space
- permeability is connected flow capacity

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Igneous rocks record the cooling history and composition of magma or lava: slow intrusive cooling favours coarse crystals, while rapid extrusive cooling favours fine, glassy or vesicular textures.

02

STAGE 02 FROM CONTINENTAL DRIFT TO PLATE TECTONICS AND WILSON-CYCLE CLOSURE

FROM CONTINENTAL DRIFT TO PLATE TECTONICS AND WILSON-CYCLE CLOSURE · WEGENER'S DRIFT CLAIM · SHELF-EDGE FIT · MATCHING ROCKS AND FOSSILS · LATER PALAEOMAGNETIC SUPPORT LIMITATION · NO ADEQUATE DRIVING MECHANISM · SEA-FLOOR SPREADING RIDGE CREATION · MAGNETIC STRIPES

WEGENER'S DRIFT CLAIM

shelf-edge fit → matching rocks and fossils → palaeoclimate → later palaeomagnetic support limitation: no adequate driving mechanism → SEA-FLOOR SPREADING ridge creation → magnetic stripes → older/deeper floor outward trench destruction

CORE CLAIM
- WEGENER'S DRIFT CLAIM
- shelf-edge fit → matching rocks and fossils
- palaeoclimate → later palaeomagnetic support limitation: no adequate driving mechanism

MECHANISM
- SEA FLOOR SPREADING ridge creation → magnetic stripes → older/deeper floor outward trench destruction
- PLATE TECTONICS divergence subduction transform creates crust destroys oceanic conserves crust ridge/rift + trench + arc
- shallow quakes rift → narrow sea → ocean → subduction → collision → suture

CONSEQUENCE / LIMIT
- Wilson cycle is an analogy, not a compulsory destiny for every ocean.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Plate boundaries are diagnosed by motion and products: divergence creates crust, subduction destroys oceanic crust, continent collision thickens crust, and transform motion conserves crust while generating shallow earthquakes.

03

STAGE 03 INDIAN PLATE JOURNEY AND THE THREE-PART GEOLOGICAL SKELETON

INDIAN PLATE JOURNEY AND THE THREE-PART GEOLOGICAL SKELETON · GONDWANA RIFTING · NORTHWARD INDIAN PLATE DRIFT · TETHYS CLOSURE · INDIA-EURASIA COLLISION ANCIENT PENINSULA HIMALAYAN FOLD-THRUST BELT FORELAND BASIN · NORTH-SOUTH EXAM CROSS-SECTION · SIWALIK HIMALAYA · FORELAND ALLUVIUM

Gondwana rifting → northward Indian Plate drift → Tethys closure → India-Eurasia collision ancient Peninsula Himalayan fold-thrust belt foreland basin cratons + belts → shortening + thickening flexure + sediment stable framework continuing uplift/seismicity Indo-Ganga-Brahmaputra → NORTH-SOUTH EXAM CROSS-SECTION → Tethyan / Greater / Lesser / Siwalik Himalaya → foreland alluvium

CORE CLAIM
- Gondwana rifting → northward Indian Plate drift → Tethys closure
- India-Eurasia collision ancient Peninsula Himalayan fold-thrust belt foreland basin cratons + belts
- shortening + thickening flexure + sediment stable framework continuing uplift/seismicity Indo-Ganga-Brahmaputra

MECHANISM
- NORTH-SOUTH EXAM CROSS-SECTION
- Tethyan / Greater / Lesser / Siwalik Himalaya → foreland alluvium
- Peninsular cratons, rifts, basins and Deccan cover

CONSEQUENCE / LIMIT
- EVIDENCE LADDER superposition → cross-cutting → unconformity → fossil correlation
- radiometric event age → palaeomagnetism → geophysics
- Stable Peninsula means relatively stable, not uniform, flat or aseismic.

MECHANISM / VERDICT — Stable Peninsula means relatively stable, not uniform, flat or aseismic.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: India's geological skeleton consists of an ancient Peninsular shield, an active Himalayan fold-thrust belt and a sediment-filled Indo-Ganga-Brahmaputra foreland basin linked by collision, flexure, erosion and deposition.

04

STAGE 04 INDIAN ROCK SYSTEMS, BASINS AND RESOURCE ASSOCIATIONS

INDIAN ROCK SYSTEMS, BASINS AND RESOURCE ASSOCIATIONS · INDIAN EXPRESSION · SAFE ASSOCIATION · GNEISS + SUPRACRUSTALS · PROCESS-SPECIFIC ORE · CUDDAPAH, VINDHYAN · STONE BOUNDED ORES · FAULTED INLAND BASINS

INDIAN EXPRESSION → SAFE ASSOCIATION → gneiss + supracrustals → process-specific ore → Cuddapah, Vindhyan

CORE CLAIM
- INDIAN EXPRESSION
- SAFE ASSOCIATION
- gneiss + supracrustals
- process-specific ore
- Cuddapah, Vindhyan

MECHANISM
- coal, not universal
- Marine basins
- coasts and offshore
- petroleum potential
- Deccan Trap

CONSEQUENCE / LIMIT
- GONDWANA BASINS: Damodar
- PETROLEUM SYSTEM
- source → maturation → migration → reservoir → seal → trap → timing
- DECCAN ARCHITECTURE
- flow top / vesicles / fractures / weathered contact / dense flow interior

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: source → maturation → migration → reservoir → seal → trap → timing.

05

STAGE 05 HIMALAYA, FORELAND, COASTS, ISLANDS AND STRUCTURAL CONTROLS

HIMALAYA, FORELAND, COASTS, ISLANDS AND STRUCTURAL CONTROLS · HIMALAYA COLLISION OROGEN · SUTURE FRAMEWORK · FORELAND FLEXURE AND FILL · BHANGAR/KHADAR ALLUVIAL EXPRESSIONS · PENINSULAR STRUCTURES · COASTS AND ISLANDS · NARMADA-SON AND TAPI RIFTS

HIMALAYA collision orogen: HFT/MFT → MBT → MCT → suture framework

FORELAND FLEXURE AND FILL → Bhabar → Terai → bhangar/khadar alluvial expressions → PENINSULAR STRUCTURES

CORE CLAIM
- HIMALAYA collision orogen: HFT/MFT → MBT → MCT → suture framework
- FORELAND FLEXURE AND FILL
- Bhabar → Terai → bhangar/khadar alluvial expressions
- PENINSULAR STRUCTURES
- COASTS AND ISLANDS

MECHANISM
- west: rifted-margin sectors
- Cambay and Kachchh
- east: deltaic sediment basins
- Godavari rift basin
- Andaman: active island arc joints, faults, bedding

CONSEQUENCE / LIMIT
- + erosion + base level + human action modify expression
- Rift + extensional zone
- graben = down-dropped block
- Lineament = mapped linear feature, not automatically an active fault.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: A rift is an extensional crustal zone, a graben is a down-dropped fault block, and a lineament is a mapped linear feature that is not automatically an active fault.

06

STAGE 06 INDIA SEISMIC EXPRESSION AND GEOLOGY-TO-RISK TRANSLATION

INDIA SEISMIC EXPRESSION AND GEOLOGY-TO-RISK TRANSLATION · TECTONIC AND INHERITED STRUCTURE · NE ANDAMAN ARC KACHCHH · PENINSULA ACTIVE COLLISION SUBDUCTION INHERITED FAULTS/RIFTS STRONG SHAKING QUAKE · SITE FILTER · ALLUVIUM, BASIN DEPTH, GROUNDWATER, SLOPE, CONSTRUCTION · EXPOSURE + VULNERABILITY - CAPACITY · DISASTER RISK

TECTONIC AND INHERITED STRUCTURE

Himalaya / NE Andaman arc Kachchh / Peninsula active collision → subduction inherited faults/rifts strong shaking quake + tsunami intraplate reactivation → SITE FILTER: alluvium, basin depth, groundwater, slope, construction

CORE CLAIM
- TECTONIC AND INHERITED STRUCTURE
- Himalaya / NE Andaman arc Kachchh / Peninsula active collision
- subduction inherited faults/rifts strong shaking quake + tsunami intraplate reactivation

MECHANISM
- SITE FILTER: alluvium, basin depth, groundwater, slope, construction
- EXPOSURE + VULNERABILITY - CAPACITY
- DISASTER RISK

CONSEQUENCE / LIMIT
- BIS Zones II-V are broad design generalisations, not predictions.
- Geological association must be followed by site evidence and engineering.
- RESPONSE: map → investigate → code → inspect → retrofit → prepare.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: India's seismicity reflects active Himalayan and Andaman plate boundaries plus reactivation of inherited structures such as Kachchh and selected Peninsular faults; seismic zoning is a design generalisation, not earthquake prediction.

07

STAGE 07 MAP-ANSWER AND MAINS SPINE FOR INDIA'S GEOLOGICAL ARCHITECTURE

MAP-ANSWER AND MAINS SPINE FOR INDIA'S GEOLOGICAL ARCHITECTURE · HOW DOES GEOLOGICAL STRUCTURE SHAPE INDIA'S SURFACE GEOGRAPHY? · 1. DEFINE · AGE + COMPOSITION + ROCK ARCHITECTURE + TECTONIC STRUCTURES · 2. DRAW · FORELAND PLAIN · PENINSULAR SHIELD CROSS-SECTION · 3. SEQUENCE

QUESTION: How does geological structure shape India's surface geography?

1. DEFINE: age + composition + rock architecture + tectonic structures → 2. DRAW: Himalaya → foreland plain → Peninsular shield cross-section → 3. SEQUENCE: Gondwana rift → drift → collision → uplift → sediment fill

CORE CLAIM
- QUESTION: How does geological structure shape India's surface geography?
- 1. DEFINE: age + composition + rock architecture + tectonic structures
- 2. DRAW: Himalaya → foreland plain → Peninsular shield cross-section
- 3. SEQUENCE: Gondwana rift → drift → collision → uplift → sediment fill

MECHANISM
- 4. NAMED UNITS cratons
- Purana basins
- Gondwana basins
- 5. LINK PROCESS TO CONSEQUENCE structure → relief/drainage/water/resource/seismic expression

CONSEQUENCE / LIMIT
- 6. QUALITY associations = not reserves; stable is not aseismic; alluvium has structure
- 7. VERDICT inherited geology is filtered through tectonics, climate, erosion and use
- Answer order: thesis → map → evidence → mechanism → India example
- limitation → graded conclusion.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 7. VERDICT inherited geology is filtered through tectonics, climate, erosion and use.

08

STAGE 08 EVIDENCE LADDER AND CONFIDENCE CONTROL

EVIDENCE LADDER AND CONFIDENCE CONTROL · EVIDENCE LADDER · 1. ARCHAEOLOGY GIVES CONTEXT; INSCRIPTIONS ACTS, COINS CIRCULATION; TEXTS · 2. DATE EACH LAYER, SAMPLE OR RECORD BEFORE COMPARING · 3. TRIANGULATE INDEPENDENT EVIDENCE ONLY AT MATCHING TIME, PLACE · FIT THE SOURCE TO THE QUESTION AND STATE THE

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
EVIDENCE LADDER	2. Date each layer, sample or record before comparing sources.	VERDICT: Fit the source to the question and state the residual uncertainty.
1. Archaeology gives context; inscriptions acts; coins circulation; texts intent.	3. Triangulate independent evidence only at matching time, place and scale.	EVIDENCE LADDER

CORE CLAIM
- EVIDENCE LADDER
- 1. Archaeology gives context; inscriptions acts; coins circulation; texts intent.

MECHANISM
- 2. Date each layer, sample or record before comparing sources.
- 3. Triangulate independent evidence only at matching time, place and scale.

CONSEQUENCE / LIMIT
- VERDICT: Fit the source to the question and state the residual uncertainty.
- EVIDENCE LADDER

MECHANISM / VERDICT — VERDICT: Fit the source to the question and state the residual uncertainty.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Fit the source to the question and state the residual uncertainty.

09

STAGE 09 EXAMINER TRAPS AND CONTESTED BOUNDARIES

EXAMINER TRAPS AND CONTESTED BOUNDARIES · CLOSE DISTINCTIONS · 1. TEXTS ARE LAYERED; INSCRIPTIONS FORMULAIC; COINS TRAVEL BEYOND · 2. SCIENTIFIC DATES ARE SAMPLE RANGES; MATERIAL CULTURE IS · 3. ABSENCE MATTERS ONLY AFTER SURVIVABILITY, SEARCH COVERAGE AND · SOURCE PROXIMITY IMPROVES EVIDENCE, NOT AUTOMATIC NEUTRALITY. · CLOSE DISTINCTION · CRUST IS COMPOSITIONAL WHILE LITHOSPHERE IS MECHANICAL; CONTINENTAL CRUST

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
CLOSE DISTINCTIONS	3. Absence matters only after survivability, search coverage and patterned silence.	continental crust is generally granitic and less dense, oceanic crust basaltic and denser, and a shield, craton, platform,
1. Texts are layered; inscriptions formulaic; coins travel beyond political borders.	VERDICT: Source proximity improves evidence, not automatic neutrality.	
2. Scientific dates are sample ranges; material culture is not language or ethnicity.	CLOSE DISTINCTION: Crust is compositional while lithosphere is mechanical	

CORE CLAIM
- CLOSE DISTINCTIONS
- 1. Texts are layered; inscriptions formulaic; coins travel beyond political borders.
- 2. Scientific dates are sample ranges; material culture is not language or ethnicity.

MECHANISM
- 3. Absence matters only after survivability, search coverage and patterned silence.
- VERDICT: Source proximity improves evidence, not automatic neutrality.
- CLOSE DISTINCTION: Crust is compositional while lithosphere is mechanical

CONSEQUENCE / LIMIT
- continental crust is generally granitic and less dense, oceanic crust basaltic and denser, and a shield, craton, platform,

MECHANISM / VERDICT — VERDICT: Source proximity improves evidence, not automatic neutrality.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Sedimentary rocks convert weathered, transported or chemically and biologically produced material into layered rock through deposition, compaction and cementation, while preserving clues such as fossils, ripples and mud cracks.

10

STAGE 10 INTEGRATED ANSWER SPINE AND QUALIFIED CONCLUSION

INTEGRATED ANSWER SPINE AND QUALIFIED CONCLUSION · ANSWER SPINE · 1. DECODE THE DIRECTIVE AND SELECT THE SOURCE CLASS · 2. WRITE CLAIM · NAMED EVIDENCE · 3. USE 2018 TRAVELLERS, 2023 LITERARY AUTHORS AND 2024 · INDEPENDENT CONVERGENCE STRENGTHENS HISTORY WITHOUT ERASING DIFFERENCES. · EVIDENCE LIMIT

ANSWER SPINE

1. Decode the directive and select the source class that fits the claim. → 2. Write claim → named evidence → inference → limitation → corroboration. → 3. Use 2018 travellers, 2023 literary authors and 2024 site matching as anchors.

CORE CLAIM
- ANSWER SPINE
- 1. Decode the directive and select the source class that fits the claim.
- 2. Write claim → named evidence → inference → limitation → corroboration.

MECHANISM
- 3. Use 2018 travellers, 2023 literary authors and 2024 site matching as anchors.
- VERDICT: Independent convergence strengthens history without erasing differences.
- EVIDENCE LIMIT: Present relief is a multi-stage tectonic-denudational product

CONSEQUENCE / LIMIT
- rock age, orogeny, mineralisation and surface form need not coincide, so geological association supports probability rather than deterministic

MECHANISM / VERDICT — VERDICT: Independent convergence strengthens history without erasing differences.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: India's surface geography is best explained as inherited geological structure filtered through tectonics, weathering, climate, erosion, sediment transfer and human use, so every high-scoring answer must connect evidence, mechanism, consequence and limitation.

11

STAGE 11 EVIDENCE LADDER AND CONFIDENCE CONTROL — DEEP-REVIEW SYNTHESIS 2

EVIDENCE LADDER AND CONFIDENCE CONTROL — DEEP-REVIEW SYNTHESIS 2 · EVIDENCE LADDER · FROM SOURCE BOOK · STANDARD KNOWLEDGE · CURRENT AFFAIRS. · 2. SEDIMENTARY CHARACTERISTICS · DERIVED FROM PRE-EXISTING ROCK IGNEOUS · 3. QUARTZITE SANDSTONE HARD; SOURCE ROCK EASILY IDENTIFIED

EVIDENCE LADDER → 1. + from source book + inference / standard knowledge = current affairs. → 2. Sedimentary characteristics: derived from pre-existing rock igneous → 3. Quartzite Sandstone Hard; source rock easily identified → VERDICT: Source type controls the strength and limits of the historical claim.

CORE CLAIM
- EVIDENCE LADDER
- 1. + from source book + inference / standard knowledge = current affairs.

MECHANISM
- 2. Sedimentary characteristics: derived from pre-existing rock igneous...
- 3. Quartzite Sandstone Hard; source rock easily identified

CONSEQUENCE / LIMIT
- VERDICT: Source type controls the strength and limits of the historical claim.
- EVIDENCE LADDER

MECHANISM / VERDICT — VERDICT: Source type controls the strength and limits of the historical claim.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Source type controls the strength and limits of the historical claim.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · 47. OLDER INDIAN ROCK-SYSTEM SCHEMES · USEFUL ONLY WHEN DECLARED · 48. FIELD IDENTIFICATION AND METAMORPHIC NUANCE · 49. CRATONS, MOBILE BELTS AND ORE SYSTEMS · 50. BASIN ARCHITECTURE BEYOND THE NAME LIST · 51. DECCAN HYDROGEOLOGY AND LANDSCAPE REFINEMENT · 52. HIMALAYAN STRUCTURAL COMPLEXITY BEYOND A CLASSROOM SECTION.

OPTIONAL SOURCE DEPTH
- 47. Older Indian rock-system schemes useful only when declared
- 48. Field identification and metamorphic nuance
- 49. Cratons, mobile belts and ore systems
- 50. Basin architecture beyond the name list

ADVANCED DEBATE
- 51. Deccan hydrogeology and landscape refinement
- 52. Himalayan structural complexity beyond a classroom section
- 53. Foreland, sediment routing and basin response
- 54. Hard-rock aquifers and engineering investigations

LEGACY / NUANCE
- 55. Mineral security without geological overclaim
- 56. Map and answer architecture
- Granite, diorite and gabbro provide coarse intrusive end-members; rhyolite, andesite and basalt are broad extrusive partners.
- Slate cleavage, schistosity and gneissic banding are fabrics, not sedimentary beds.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels